Practical no. 11

MICROCONTROLLER & EMBEDDED SYSTEMS LABORATORY MANUAL

Experiment: Interfacing R307/R305 Fingerprint Sensor & I2C 16×2 LCD

Target Platform: Raspberry Pi 4 Model B (Linux / Python 3) **Protocols:** Hardware UART & I2C Bus Interface

1. AIM & OBJECTIVES

To interface an optical fingerprint sensor (R307/R305) via hardware UART communication (`/dev/serial0`) and a 16×2 character LCD using a PCF8574 (HW-61) I2C interface adapter with Raspberry Pi 4, register fingerprint biometric templates into flash memory, and perform real-time 1:N verification with status display messages.

2. HARDWARE COMPONENTS & VISUAL REFERENCE



Figure 1: 16×2 Character LCD with Soldered HW-61 (PCF8574) I2C Backpack



Figure 2: R307 / R305 Optical Fingerprint Sensor Module

3. PINOUT & INTERFACING CONNECTION TABLE

Device / Module	Module Pin	Pin Function	Raspberry Pi Pin Name	Physical Pin No.
16×2 I2C LCD (HW-61 Backpack)	**GND**	Ground Reference	GND	Pin 6
	VCC	Supply Voltage (+5V)	5V Power	Pin 2
	SDA	I2C Serial Data	GPIO 2 (I2C1 SDA)	Pin 3
	SCL	I2C Serial Clock	GPIO 3 (I2C1 SCL)	Pin 5
R307 / R305 Fingerprint Sensor	**Red (VCC)**	Module Power (+5V)	5V Power	Pin 4
	Black (GND)	Ground Reference	GND	Pin 9 / 14
	Yellow/Green (TXD)	UART Transmit Data	GPIO 15 (UART0 RXD)	Pin 10
	White/Blue (RXD)	UART Receive Data	GPIO 14 (UART0 TXD)	Pin 8

4. SYSTEM CONFIGURATION & DRIVER INSTALLATION

1. **Enable Interfaces:** Run `sudo raspi-config` . Under *Interface Options*, enable **I2C**. Under **Serial Port**, select **No** for login shell and **Yes** for hardware serial. Reboot with `sudo reboot` .
2. **Install Dependencies:**

```
sudo apt-get update && sudo apt-get install -y i2c-tools python3-pip python3-serial
pip3 install RPLCD adafruit-circuitpython-fingerprint smbus2
```

3. **Verify I2C Address:** Run `i2cdetect -y 1` (Address defaults to `0x27` or `0x3f`).

5. PYTHON SOURCE PROGRAMS

A. Enrollment Script (enroll.py)

```
import serial
import adafruit_fingerprint

uart = serial.Serial("/dev/serial0", baudrate=57600, timeout=1)
finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)

location = int(input("Enter Slot ID (1-127): "))

print("Place finger on sensor...")

while finger.get_image() != adafruit_fingerprint.OK:
    pass

finger.image_2_tz(1)

print("Remove finger...")
time.sleep(2)

while finger.get_image() == adafruit_fingerprint.OK:
    pass

print("Place SAME finger again...")

while finger.get_image() != adafruit_fingerprint.OK:
    pass

finger.image_2_tz(2)

if finger.create_model() == adafruit_fingerprint.OK:
    if finger.store_model(location) == adafruit_fingerprint.OK:
        print(f"Enrolled successfully at Slot {location}!")
    else:
        print("Error saving template.")
else:
    print("Fingerprint mismatch. Try again.")
```

B. Verification System (`main_verify.py`)

```
import time
import serial
from RPLCD.i2c import CharLCD
import adafruit_fingerprint
# LCD setup
lcd = CharLCD('PCF8574', 0x27, cols=16, rows=2)
# Fingerprint sensor setup
uart = serial.Serial(
    "/dev/serial0",
    baudrate=57600,
    timeout=1
)
finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)
def show(l1, l2=""):
    lcd.clear()

    lcd.cursor_pos = (0, 0)
    lcd.write_string(l1[:16])

    lcd.cursor_pos = (1, 0)
    lcd.write_string(l2[:16])
# Startup message
show("System Ready", "Place Finger")
while True:
    # Wait for finger
    if finger.get_image() == adafruit_fingerprint.OK:
        show("Processing...", "Please Wait")
        # Convert fingerprint image
        if finger.image_2_tz(1) == adafruit_fingerprint.OK:
            # Search stored fingerprints
            if finger.finger_search() == adafruit_fingerprint.OK:
                # Fingerprint matched
                show(
                    "FINGER MATCHED",
                    f"ID: {finger.finger_id}"
                )
                time.sleep(3)

            else:
                # Fingerprint not found
                show("FINGERPRINT", "UNMATCHED")
                time.sleep(1.5)

                show("Access Denied", "")
                time.sleep(2)

        else:
            show("Image Error", "")
            time.sleep(2)

        # Ready for next finger
        show("System Ready", "Place Finger")

    time.sleep(0.1)
```

6. STEP-BY-STEP PRACTICAL EXECUTION

1. Power off the Raspberry Pi 4 and wire all modules according to the pin connection table.
2. Power on the Raspberry Pi and confirm I2C bus detection using `i2cdetect -y 1`.
3. Run `python3 enroll.py` in the terminal and follow on-screen prompts to register sample fingerprints into Slot ID 1, 2, etc.
4. Launch the verification terminal: `python3 main_verify.py`.
5. Test with enrolled fingers (verifying "RFID UID MATCHED") and unenrolled fingers (verifying "RFID UID UNMATCHED" & "Access Denied").

7. TROUBLESHOOTING & COMMON ERRORS

LCD Contrast Tuning: If the LCD backlight illuminates but text is not visible or black boxes appear, adjust the blue trim potentiometer on the back of the HW-61 I2C adapter using a screwdriver.

UART Tx/Rx Cross Connection: Ensure UART lines are cross-connected: Sensor TXD connects to Pi RXD (Physical Pin 10), and Sensor RXD connects to Pi TXD (Physical Pin 8).

8. RESULT

The R307/R305 optical fingerprint sensor and I2C 16×2 character LCD were successfully interfaced with the Raspberry Pi 4. Biometric template enrollment, 1:N fingerprint authentication matching, and real-time visual output updates on the LCD were validated.